a_compact_commercial_air-source_heat_pump

Comprehensive Hardware Design Analysis

Revision 1 | Generated 04/05/2026

Executive Summary

Generated

Project ID	**a_compact_commercial_airsource_heat_pump**
Industry Domain	**hvac_commercial**
Engine Version	**PDF Engine v2.1 (Full Detail)**
Target Quantity	**Not specified**

ARCHITECTURE AT A GLANCE

MODULES	BILL OF MATERIALS ROWS	IDENTIFIED RISKS
9	**0**	**25**

ECONOMICS

EST. UNIT COST	TARGET CEILING	HEADROOM
£0.00	**None**	**N/A**

TOTAL NON-RECURRING ENGINEERING	UNIT COST PLUS AMORTISED NON-RECURRING ENGINEERING
£0.00	**£0.00**

Contents

1. Brief & Requirements

1.1 Overview & Context

Subject / Use Case

Direct replacement of aging 30kW-50kW gas commercial boilers, providing both ambient space heating and domestic hot water generation via an internal hydrobox component.

Mission Statement

To accelerate the decarbonization of small commercial buildings in the UK with a high-efficiency, highly affordable R290 heat pump solution.

Target Customers

Small office buildings, boutique retail spaces, cafes, light industrial units, and small schools in the UK.

Why Now (Market Timing)

The UK is aggressively phasing out commercial gas boilers to meet Net Zero 2050 targets, while F-Gas regulations are simultaneously forcing a transition from R410A/R32 to natural refrigerants like R290.

Full Synthesis Report

This research analyzes the feasibility and market dynamics of developing a 30kW compact commercial air-source heat pump (ASHP) utilizing R290 (propane) refrigerant, specifically designed for small commercial buildings in the UK. The specified target of a COP >= 3.5 at A2/W35 is technically completely viable and highly competitive, achievable through the integration of a modern R290-optimized scroll compressor, an axial fan with ECM motor, and a highly efficient stainless steel brazed plate heat exchanger. R290 presents a near-zero Global Warming Potential (GWP of 3), positioning the product perfectly in an era of stringent UK F-Gas phase-downs. However, the architectural brief specifying a 'split system' (indoor plus outdoor unit) using R290 introduces significant engineering and regulatory hurdles. Due to BS EN 378 safety standards governing highly flammable (A3) refrigerants, introducing an R290 refrigerant line into an indoor unit requires extreme charge limits or explosive-proof ATEX-rated indoor enclosures. The standard industry pivot is to utilize a monobloc architecture or a 'hydro-split' where the R290 circuit is sealed within the outdoor unit and only warm water/glycol is pumped indoors to the control/hydrobox. Financially, the unit cost target of £4,500 at a low production volume of 500 units per year is highly aggressive. Typical premium commercial 30kW units cost between £6,500 and £9,000

Data Sources:
[User] Founder brief text
[LLM] Gemini 3.1 Pro — research synthesis

to manufacture at low volumes. Achieving the £4,500 target will require rigorous value engineering, strategic sourcing of commercial off-the-shelf (COTS) scroll compressors and fans, and a heavily standardized sheet metal chassis. Attaining MCS certification is non-negotiable for UK market penetration, as it is a prerequisite for government capital grants (like the Boiler Upgrade Scheme, where applicable or equivalent commercial tax benefits) and favorable financing.

Data Sources:
[User] Founder brief text
[LLM] Gemini 3.1 Pro — research synthesis

1.2 Market & Competitors

Competitor Landscape

Clivet

Product	**Edge EVO 2.0 EXCEL (Commercial Monobloc/Cascade)**
Pricing Strategy	**Approx. £7,500 - £8,500 (unit cost via distributors)**
Technical Specs	**30kW cascaded output (R32 moving to R290 variants), scroll compressor, DC inverter axial fans.**

STRENGTHS

Highly scalable with native cascading controllers, strong commercial HVAC reputation.

WEAKNESSES

Currently transitioning portfolios to R290, high component cost, bulky outdoor footprint.

DIFFERENTIATION ANGLE

Our target aims to undercut their pricing by 40% utilizing a simplified core chassis while maintaining the 30kW single unit format with R290.

Viessmann

Product	**Vitocal 250-A Pro**
Pricing Strategy	**Approx. £11,000+**
Technical Specs	**To 40kW output, R290 refrigerant, COP >3.5, advanced indoor control unit.**

STRENGTHS

Premium build quality, extremely quiet acoustically, highly trusted brand in the UK commercial segment.

WEAKNESSES

Very expensive, oversized for end-users highly sensitive to capital expenditure (CapEx).

DIFFERENTIATION ANGLE

Stripped down 'value-engineered' design focusing purely on core performance and affordability (£4,500 target), making ROI significantly faster for small businesses.

Data Sources:
[User] Founder brief text
[LLM] Gemini 3.1 Pro — research synthesis

Vaillant

Product	**aroTHERM plus (Cascaded setup for 30kW)**
Pricing Strategy	**Approx. £9,000 for twin units + cascade controller**
Technical Specs	**2 x 15kW units cascaded, R290 monobloc, SCOP up to 4.5.**

STRENGTHS

Excellent MCS compliance, very high SCOP, readily available supply chain.

WEAKNESSES

Requires purchasing two separate 15kW units to reach the 30kW demand, doubling installation footprint and increasing piping complexity.

DIFFERENTIATION ANGLE

Providing a single, compact 30kW system to minimize external footprint and installation time compared to cascaded domestic units.

Data Sources:
[User] Founder brief text
[LLM] Gemini 3.1 Pro — research synthesis

1.3 Engineering Constraints

Core Targets

Unit Cost Ceiling	**£0.00**
Max Mass	**-**
Target Process	
Target Material	
Tolerance Target	
Production Quantity	

Research Sources

TITLE	TYPE	YEAR	RELEVANCE
BSRIA UK Commercial Heat Pump Market Report 2023	Market Research	2026	Provides volumetric, pricing, and adoption trend data for the 20-50kW commercial heat pump segment in the UK.
MCS MIS 3005-I: Heat Pump Standard	Regulatory Standard	2026	Outlines the installation and product efficiency requirements necessary to achieve MCS certification in the UK.
BS EN 378: Refrigerating systems and heat pumps. Safety and environmental requirements	Engineering Standard	2026	Critically dictates the maximum allowable charge limits and safety requirements for R290 (propane) indoor systems.
Emerson Copeland R290 Commercial Scroll Compressor Guidelines	Technical Datasheet	2026	Details the performance envelopes, COP modeling, and safety handling requirements for R290 scroll compressors at 30kW capacities.
UK F-Gas Regulations & Phase-down Schedule	Government Policy	2026	Mandates the shift away from high GWP refrigerants, validating the long-term strategic choice of R290.

Data Sources:
[User] Founder brief text
[LLM] Gemini 3.1 Pro — research synthesis

2.1 MIS-3005-I

Microgeneration Certification Scheme (MCS)

Status: not-started

Owner: Compliance Engineer

Standard Summary

UK standard for microgeneration products determining quality, efficiency, and accurate performance ratings.

Applicability Assessment

> Both indoor and outdoor units to qualify for UK grants.

Engineering Design Impact

Requires rigorous third-party COP testing, noise emission compliance, and extensive user documentation.

EVIDENCE REQUIRED

Test reports from a UKAS accredited laboratory verifying testing to BS EN 14511.

GAP ACTION

Source a UKAS-accredited lab for BS EN 14511 testing and initiate MCS manufacturer application.

Data Sources:

[LLM] Gemini 3.1 Pro — standards extraction

2.2 BS-EN-378

Refrigerating systems - Safety

Status: not-started | **Owner: Mechanical Design Lead**

Standard Summary

Safety and environmental standard detailing design and charge limits for refrigerants (specifically A3 flammability).

Applicability Assessment

> Entire R290 refrigerant circuit, with severe impact if lines enter the indoors.

Engineering Design Impact

Must ensure R290 charge limits are not exceeded for the indoor volume, highly suggesting a 'hydro-split' rather than direct expansion indoors.

EVIDENCE REQUIRED

FMEA, ATEX component certificates (if indoors), and refrigerant charge calculation logs.

GAP ACTION

Evaluate split-system architecture to contain the entire R290 primary loop within the outdoor unit.

Data Sources:
[LLM] Gemini 3.1 Pro — standards extraction

2.3 BS-EN-60335-2-40

Electrical Safety for Heat Pumps

Status: not-started Owner: Electrical Engineer

Standard Summary

Specific electrical safety standards for electrical heat pumps, air-conditioners, and dehumidifiers.

Applicability Assessment

> All electrical components: scroll compressor, axial fan, hydrobox pumps, and controllers.

Engineering Design Impact

Mandates specific earth bonding, IP ratings for outdoor use, and safety shut-offs for the compressor motor.

EVIDENCE REQUIRED

CB test certificate and electrical schematic safety review.

GAP ACTION

Conduct a pre-compliance electrical safety test on the first prototype.

Data Sources:
[LLM] Gemini 3.1 Pro — standards extraction

2.4 PESR-2016

Pressure Equipment (Safety) Regulations

Status: not-started | Owner: Manufacturing Engineer

Standard Summary

UK equivalent to the EU PED, strictly regulating systems working under pressure.

Applicability Assessment

> Stainless steel heat exchanger, compressor casting, and refrigerant piping.

Engineering Design Impact

Requires burst-pressure testing and specific welding qualifications for the copper/stainless joints.

EVIDENCE REQUIRED

Material certificates, welding schedules, and hydrostatic burst test reports.

GAP ACTION

Ensure heat exchanger vendor provides UKCA/PESR compliant burst testing data.

Data Sources:
[LLM] Gemini 3.1 Pro — standards extraction

2.5 ErP-Lot-1

Energy Related Products Directive (Ecodesign)

Status: not-started | Owner: Product Manager

Standard Summary

Minimum energy performance standards and seasonal space heating energy efficiency tagging.

Applicability Assessment

> Overall heat pump system marketing and sale in the UK/EU.

Engineering Design Impact

Must meet minimum SCOP requirements and display appropriate energy labels (e.g., A++ or A+++).

EVIDENCE REQUIRED

SCOP test reports at various climate loads and acoustic noise power level data.

GAP ACTION

Create SCOP modeling using compressor maps to baseline expected ErP ratings.

Data Sources:
[LLM] Gemini 3.1 Pro — standards extraction

3. Sizing & Spatial Optimisation

INFEASIBLE DESIGN ALLOCATION
The current sizing constraints are not satisfied. Module requirements exceed the available envelope.

System Envelope

KIND	RULES DOMAIN	FEASIBLE
generic_room	**hvac_commercial**	**No**

Internal Dimensions (W×D×H)	**5000 × 5000 × 3000 mm**
Total Internal Floor Area	**25 m²**
Total Internal Volume	**75 m³**
Available Floor Budget	**21.25 m²**

Module Dimension Allocations

MODULE	W × D × H (MM)	AREA M²	MOUNT	SCALE
compressor_assembly	2236 × 2236 × 2400	5	floor	mass_proportional
evaporator_coil_module	2236 × 2236 × 2400	5	floor	mass_proportional
air_movement_module	2236 × 2236 × 2400	5	floor	mass_proportional
condenser_bphe_module	2236 × 2236 × 2400	5	floor	mass_proportional
refrigerant_management_skid	2236 × 2236 × 2400	5	floor	mass_proportional

Data Sources:
[Deterministic] Sizing solver and envelope constraints

outdoor_electrical_enclosure	2236 × 2236 × 2400	5	floor	mass_proportional
structural_chassis	2236 × 2236 × 2400	5	floor	mass_proportional
indoor_hydrobox_module	2236 × 2236 × 2400	5	floor	mass_proportional
control_gateway_hmi	2236 × 2236 × 2400	5	floor	mass_proportional

Identified Conflicts

- Total module footprint (45.0m2) exceeds envelope budget (21.3m2)

Optimisation Recommendations

- Consider a larger envelope
- Optimize module footprint

Data Sources:
[Deterministic] Sizing solver and envelope constraints

4.1 Compressor Assembly

Lead Time: 12 weeks | Status: draft

Module Purpose

Compresses low-pressure R290 gas into a high-pressure, high-temperature gas to drive the thermodynamic heating cycle.

Why it matters to the system

> It is the primary driver of system efficiency and the single most expensive component; sizing and sourcing it correctly is the only way to hit the aggressive £4,500 target cost.

Detailed Technical Description

The compressor assembly represents the thermodynamic heart of the heat pump. Utilizing a commercial off-the-shelf (COTS) DC inverter scroll compressor optimized for R290, it intakes cool, low-pressure propane gas from the evaporator and mechanically compresses it. This compression drastically increases the kinetic energy and temperature of the gas, preparing it for heat exchange to the hydronic loop. Because R290 is an A3 (highly flammable) refrigerant, the compressor features specialized spark-proof electrical terminals (ATEX rated) and a sealed casing to mitigate ignition risks.

From a physical perspective, the scroll mechanism allows for smooth, continuous compression, achieving the demanding target COP of 3.5 at A2/W35. The inclusion of an inverter allows the compressor to modulate its speed (capacity) based on thermal demand, which is crucial for maximizing seasonal efficiency (SCOP) in the variable UK climate.

Acoustics and vibrations are significant secondary physics issues in this module. The compressor generates substantial mechanical noise and low-frequency vibrations that must be damped using specialized rubber isolators and a heavy-duty acoustic jacket, ensuring compliance with the stringent MCS noise emission standards.

SYSTEM INPUTS	SYSTEM OUTPUTS

Data Sources:
[LLM] Gemini 3.1 Pro — module decomposition

- Low-pressure R290 vapor
- 3-Phase 400V AC power (via inverter)

- High-pressure R290 vapor
- Mechanical heat and vibration

Data Sources:

[LLM] Gemini 3.1 Pro — module decomposition

4.1.2 Specifications & Risk Profile

Expected Key Parts

- R290 Optimized DC Inverter Scroll Compressor
- ATEX-rated terminal box
- Vibration isolation mounts
- Crankcase heater
- Acoustic jacket

Identified Failure Modes

- Liquid slugging causing mechanical scroll fracture
- Inadequate lubrication leading to bearing seizure
- Terminal arc igniting a localized R290 leak

Open Questions

- Exact harmonic vibration frequencies at peak RPM
- Long-term wear rates of seals with specific commercial compressor oils

Data Sources:
[LLM] Gemini 3.1 Pro — module decomposition

4.1.3 Cost Breakdown

No bill of materials rows allocated to this module yet.

Data Sources:
[LLM] Gemini 3.1 Pro — part generation
[Deterministic] Cost model calculation

4.2 Evaporator Coil Module

Lead Time: 10 weeks | Status: draft

Module Purpose

Captures ambient environmental heat by allowing liquid R290 to boil into a gas.

Why it matters to the system

> Directly dictates the system's ability to extract heat at low ambient temperatures, making it the most critical element for achieving the target ErP and SCOP ratings in a UK winter.

Detailed Technical Description

The evaporator coil module is a sprawling fin-and-tube heat exchanger that forms the extensive 'lung' of the outdoor unit. As ambient air is drawn across the aluminum fins, heat is transferred via conduction and convection into the inner copper tubes, which carry the severely sub-cooled, low-pressure phase-change refrigerant. The hydrophilic coating on the fins ensures that condensation easily runs off, delaying frost build-up during the damp, near-freezing UK winters.

The thermodynamics are governed by the massive surface area and internal rifling of the copper tubes, which increases turbulent flow and heat transfer coefficients without excessively increasing pressure drop. R290 possesses exceptionally high volumetric capacity and thermal conductivity, allowing this commercial coil to be slightly more compact than an R410A equivalent. However, due to the tight £4,500 budget, the coil must rely on standard tube diameters and automated fin stamping to leverage economies of scale.

A major operational constraint is defrosting. When ambient temperatures are between -2°C and +4°C, frost rapidly accumulates, reducing airflow and halting heat transfer. This module must seamlessly integrate with the system's reversing valve logic to periodically flood the coil with hot R290 gas, melting the ice efficiently before returning to heating mode.

SYSTEM INPUTS	SYSTEM OUTPUTS

Data Sources:
[LLM] Gemini 3.1 Pro — module decomposition

- Ambient air (-20C to +35C)
  - Low-pressure liquid/vapor R290 mix

- Chilled exhaust air
  - Low-pressure R290 vapor

Data Sources:

[LLM] Gemini 3.1 Pro — module decomposition

4.2.2 Specifications & Risk Profile

Expected Key Parts

- Hydrophilic-coated aluminum fins
- Rifled copper tubing
- Header and distributor tubes
- Defrost temperature thermistors

Identified Failure Modes

- Micro-cracks at copper U-bends leading to total loss of R290 charge
- Fin edge damage during factory assembly or on-site
- Ice expansion fracturing the lower tube runs during incomplete defrost cycles

Open Questions

- Optimal fin pitch to balance heat transfer and frost immunity
- Exact airflow distribution characteristics across the full coil face

Data Sources:
[LLM] Gemini 3.1 Pro — module decomposition

4.2.3 Cost Breakdown

No bill of materials rows allocated to this module yet.

Data Sources:
[LLM] Gemini 3.1 Pro — part generation
[Deterministic] Cost model calculation

4.3 Air Movement Module

Lead Time: 8 weeks

Status: draft

Module Purpose

Forces massive volumes of ambient air across the evaporator coil.

Why it matters to the system

> Fan noise is the largest barrier to planning permission and MCS certification. High-efficiency EC fans are also essential for hitting ErP Lot 1 electrical efficiency targets.

Detailed Technical Description

This sub-assembly forcefully drives ambient air through the restrictive fins of the evaporator coil. To meet the 30kW demand, twin axial fans are typically utilized, driven by highly efficient EC (Electronically Commutated) motors. EC technology allows for infinitely variable speed control, meaning the fan speed can intimately track the compressor's inverter output. This not only dramatically saves parasitic electrical consumption but ensures tight control over noise levels during low-load operation.

Aerodynamically, the fan blades must overcome the static pressure drop of the tightly packed evaporator fins while minimizing turbulent boundary layer separation, which is the primary source of 'whooshing' aerodynamic noise. Designing a custom, deep drawn composite or spun sheet metal fan shroud is essential. The shroud directs airflow axially, avoiding recirculation at the blade tips, thus maximizing the fan's efficiency curve.

Noise emission is highly regulated for UK commercial installs via MCS planning constraints. The balance between air volume (required for COP) and blade tip speed (responsible for acoustics) is one of the most critical optimization loops in the entire product development cycle.

SYSTEM INPUTS	SYSTEM OUTPUTS

Data Sources:
[LLM] Gemini 3.1 Pro — module decomposition

• 230V AC Power • PWM Control Signal (0-10V)	• High volume directional airflow • Acoustic noise

Data Sources:

[LLM] Gemini 3.1 Pro — module decomposition

4.3.2 Specifications & Risk Profile

Expected Key Parts

- EC (Electronically Commutated) axial fan(s)
- Aerodynamic fan shroud
- Protective bird guard
- Motor mounting spider/cowl bracket

Identified Failure Modes

- EC motor integrated driver board failure due to transient voltage spikes
- Bearing wear leading to off-axis wobble and low-frequency hum
- Blade structural failure due to fatigue from ice-impact or resonance

Open Questions

- Exact aerodynamic stall point against heavily frosted coil
- System resonance frequencies interacting with the motor PWM frequency

Data Sources:
[LLM] Gemini 3.1 Pro — module decomposition

4.3.3 Cost Breakdown

No bill of materials rows allocated to this module yet.

Data Sources:
[LLM] Gemini 3.1 Pro — part generation
[Deterministic] Cost model calculation

4.4 Condenser BPHE Module

Lead Time: 14 weeks | Status: draft

Module Purpose

Transfers thermal energy from the high-pressure R290 gas to the hydronic (water/glycol) heating loop.

Why it matters to the system

> Limits the entire heating output and flow temperature capability; its integration isolates the flammable R290 outdoors, satisfying BS EN 378 safety mandates.

Detailed Technical Description

The condenser in this hydro-split architecture is an asymmetric Brazed Plate Heat Exchanger (BPHE). It serves as the vital boundary between the sealed A3 refrigerant circuit and the safe, building-side hydronic heating loop. Inside the unit, exceptionally hot R290 discharge gas flows through alternating corrugated stainless steel channels, while return water flows in counter-current through adjacent channels. As the R290 sheds its latent heat, it condenses from a gas back into a dense liquid.

Because R290 has tremendous thermal capacity and can safely operate at higher discharge temperatures without oil degradation, this BPHE can produce water flow temperatures up to 75°C, making it a perfect direct replacement for legacy UK commercial gas boilers. The physics of the plate corrugations are designed to induce high turbulence, scouring the boundary layer and yielding enormous heat transfer coefficients within a tiny physical footprint.

Regulatory compliance here is driven by PESR-2016 (Pressure Equipment Safety Regulations). The R290 side operates at high pressures (up to 30 bar), demanding robust copper-to-stainless brazed joints and rigorous hydrostatic burst testing. The £4,500 system cost target means specifying a standardized COTS BPHE from a tier-1 vendor rather than developing custom stamping geometries.

SYSTEM INPUTS	SYSTEM OUTPUTS

Data Sources:
[LLM] Gemini 3.1 Pro — module decomposition

- High-pressure R290 vapor
  - Return/High-temp water/glycol from building (e.g., 30°C)
- High-pressure subcooled R290 liquid
  - Flow water/glycol to building (e.g., 35°C up to 75°C)

Data Sources:

[LLM] Gemini 3.1 Pro — module decomposition

4.4.2 Specifications & Risk Profile

Expected Key Parts

- AISI 316L Brazed Plate Heat Exchanger (BPHE)
- EPDM closed-cell thermal insulation jacket
- High-pressure copper braze studs
- Stainless steel water thread connections

Identified Failure Modes

- Thermal fatigue cracking at the copper-stainless manifold braze melts
- Internal plate rupture mixing high-pressure R290 directly into the building's water pump
- Water-side fouling/scaling severely degrading heat transfer efficiency

Open Questions

- Severity of hard water scaling at 75C flow temperatures
- Precise pressure drop on the hydronic loop requiring pump re-sizing

Data Sources:

[LLM] Gemini 3.1 Pro — module decomposition

4.4.3 Cost Breakdown

No bill of materials rows allocated to this module yet.

Data Sources:
[LLM] Gemini 3.1 Pro — part generation
[Deterministic] Cost model calculation

4.5 Refrigerant Management Skid

Lead Time: 12 weeks | Status: draft

Module Purpose

Controls the directional flow, pressure drop, and purity of the R290 refrigerant loop.

Why it matters to the system

> Perfect EEV modulation is what allows the heat pump to maintain high COP across rapidly fluctuating UK weather conditions and varying thermal loads.

Detailed Technical Description

This module acts as the sophisticated circulatory control mechanism of the heat pump. The critical component is the Electronic Expansion Valve (EEV). As high-pressure liquid R290 passes through the EEV's ultra-precise stepper-motor-actuated orifice, it experiences a massive pressure drop (the Joule-Thomson effect). This forces the R290 to rapidly boil and flash into a low-temperature vapor-liquid mixture, dropping its temperature far below ambient air temperatures so it can absorb heat in the evaporator coil.

Additionally, this skid houses the heavy-duty 4-way reversing valve, which flips the entire thermodynamic cycle to perform rapid defrosting of the outdoor coil when frozen. The suction accumulator is physically positioned just before the compressor to trap any unboiled liquid refrigerant, ensuring that only pure vapor hits the compressor (preventing fatal hydraulic lock).

Due to the extreme leak-sensitivity of an A3 flammable refrigerant under BS EN 378, every joint on this skid must be brazed using strictly controlled shielding gases, with minimizing total joint count being a primary design directive for the value-engineering team. Flared connections are completely prohibited.

SYSTEM INPUTS	SYSTEM OUTPUTS

Data Sources:
[LLM] Gemini 3.1 Pro — module decomposition

- High-pressure liquid R290 (from Condenser)
- Control signals from Main PCB

- Regulated low-pressure R290 vapor mixture
- Reversed flow (during defrost)

Data Sources:

[LLM] Gemini 3.1 Pro — module decomposition

4.5.2 Specifications & Risk Profile

Expected Key Parts

- Electronic Expansion Valve (EEV)
- 4-way Reversing Valve
- Suction Accumulator / Liquid Receiver
- Filter Drier and Sight Glass
- High/Low Pressure Transducers

Identified Failure Modes

- Stepper motor failure on EEV locking the valve closed
- 4-way valve sticking in mid-cycle due to metal debris or sludge
- Micro-porosity in braze joints slowly leaking R290 charge over time

Open Questions

- Optimal PID tuning parameters for the EEV to prevent hunting
- Exact refrigerant charge weight required to balance heating and defrost modes

Data Sources:

[LLM] Gemini 3.1 Pro — module decomposition

4.5.3 Cost Breakdown

No bill of materials rows allocated to this module yet.

Data Sources:
[LLM] Gemini 3.1 Pro — part generation
[Deterministic] Cost model calculation

4.6 Outdoor Electrical Enclosure

Lead Time: 16 weeks

Status: draft

Module Purpose

Distributes high-voltage power and executes control logic while safely segregated from the flammable R290 circuit.

Why it matters to the system

> Controls every aspect of the unit's modulation and efficiency. Incorrect isolation of this box represents the primary explosive risk for the whole product.

Detailed Technical Description

The outdoor electrical enclosure is the brain and the muscle control center of the heat pump. Because the product utilizes 3-phase commercial power to drive a 30kW output, massive electrical currents are managed here. The heart of the box is the Inverter Drive, which converts 400V AC grid power into variable-frequency DC power to precisely control the compressor's RPM. Associated with this are heavy-duty EMI/RFI filters necessary to prevent electrical noise from polluting the local grid, a strict requirement under UK ENA EREC G99 directives.

The most unique design consideration for this enclosure is safety segregation. Due to the presence of R290, the entire electrical box must be hermetically isolated from the compressor and refrigerant skid. If a leak occurs, R290 gas (which is heavier than air) can pool. Therefore, this enclosure must be physically located high up in the chassis, tightly IP65 sealed, and equipped with a dedicated ventilation path to prevent explosive atmospheres from forming near its relays and contactors.

Hitting the aggressive £4,500 target requires off-the-shelf heat pump controllers (like Carel or Sanhua) with custom pre-programmed UK commercial maps, saving massive internal R&D software costs while providing guaranteed BS-EN-60335-2-40 electrical safety compliance.

Data Sources:
[LLM] Gemini 3.1 Pro — module decomposition

SYSTEM INPUTS

- 400V 3-Phase Grid Power
- Sensor Telemetry (Temp, Pressure, Flow)

SYSTEM OUTPUTS

- Modulated DC power to Compressor
- PWM to Fans/EEV
- Comms to Indoor Unit (RS485)

Data Sources:

[LLM] Gemini 3.1 Pro — module decomposition

4.6.2 Specifications & Risk Profile

Expected Key Parts

- Main system PCB / Microcontroller
- Compressor Inverter Drive
- Sealed IP65 / ATEX certified sheet metal box
- EMI/RFI mains filters
- Power contactors & Relays

Identified Failure Modes

- Inverter IGBT module burnout from inadequate thermal dissipation
- Moisture ingress causing logic PCB short circuit across high-voltage tracks
- Electromagnetic interference disrupting digital sensor readings

Open Questions

- The max heat load generated by the inverter module during continuous 30kW peak load
- Software integration timeline with off-the-shelf control board manufacturer

Data Sources:
[LLM] Gemini 3.1 Pro — module decomposition

4.6.3 Cost Breakdown

No bill of materials rows allocated to this module yet.

Data Sources:
[LLM] Gemini 3.1 Pro — part generation
[Deterministic] Cost model calculation

4.7 Structural Chassis & Casing

Lead Time: 6 weeks | Status: draft

Module Purpose

Provides the weather-resistant, load-bearing exoskeleton and acoustic barrier for the outdoor unit.

Why it matters to the system

> It is the frontline defense against the harsh UK environment, provides the necessary MCS-compliant sound-proofing, and dictates the perceived build quality by the installer.

Detailed Technical Description

The structural chassis is a heavily value-engineered sheet metal assembly that serves as the physical foundation of the unit. Supporting upwards of 280kg (mostly from the heavy twin compressors and copper coils), the base pan must possess immense torsional rigidity to survive transit and installation onto commercial rooftops or concrete plinths without warping, which could crack the rigid internal copper pipelines.

From a physics perspective, the casing plays a massive role in acoustic attenuation. Low-frequency compressor grumbles and high-frequency fan whine are trapped and absorbed by heavy-density, mass-loaded acoustic foam lining the internal panels. Furthermore, the design of the base pan must rapidly evacuate hundreds of liters of defrost condensation water away from the unit; if this water pools and freezes underneath the heat exchanger, it can destroy the lower coil runs.

To align with the £4,500 budget at low volumes (500 units/year), expensive bespoke casting or complex folded plastics are avoided in favor of CNC punched and folded EN-10346 galvanized steel. Modular panelling allows quick removal for maintenance access, an essential requirement for commercial facility managers minimizing downtime.

SYSTEM INPUTS	SYSTEM OUTPUTS

Data Sources:
[LLM] Gemini 3.1 Pro — module decomposition

- Environmental weather (Rain, snow, UV)
- Internal vibrations and weight (200kg+)

- Structural rigidity
- Acoustic damping
- Safe lifting points

Data Sources:

[LLM] Gemini 3.1 Pro — module decomposition

4.7.2 Specifications & Risk Profile

Expected Key Parts

- Galvanized steel base pan with drain channel
- Anti-corrosion powder-coated sheet metal panels
- Mass-loaded vinyl acoustic insulation
- Forklift pockets / Lifting eyes
- Hail-guard grilles

Identified Failure Modes

- Vibration fatigue tearing the sheet metal around motor mounting points
- Base pan drainage blockage causing dewatering ice heave into the coil
- Premature corrosion at spot-weld points due to coastal salt air

Open Questions

- Exact locations of resonant nodes on the exterior panels
- Tolerance stack-up issues causing panels rattle during compressor ramp-up

Data Sources:

[LLM] Gemini 3.1 Pro — module decomposition

4.7.3 Cost Breakdown

No bill of materials rows allocated to this module yet.

Data Sources:
[LLM] Gemini 3.1 Pro — part generation
[Deterministic] Cost model calculation

4.8 Indoor Hydrobox Module

Lead Time: 10 weeks | Status: draft

Module Purpose

Circulates the heated fluid through the building and provides essential hydronic system safety limitations.

Why it matters to the system

> Enables the safety of the R290 hydro-split design by moving all water distribution and backup heating indoors, protecting it from freezing and streamlining installation.

Detailed Technical Description

Because the R290 refrigerant is mandated by BS EN 378 safety limits to remain entirely outdoors, this Indoor Hydrobox Module completes the 'hydro-split' architecture. It is essentially a compact plumbing station mounted inside the commercial facility. It receives hot water from the outdoor heat exchanger and drives it forcefully through the building's heating emitters (UHF, radiators, or DHW cylinders) using a high-throughput, energy-efficient DC inverter pump.

The hydrobox manages hydronic physics: it accommodates the thermal expansion of system water via its internal diaphragm expansion vessel and protects against over-pressurization using safety relief valves. It also contains an emergency backup immersion heater (e.g., 3kW/6kW). If the outdoor unit fails, or if a severe sub-zero cold snap demands thermal output beyond the heat pump's capacity, the immersion heater kicks in to guarantee building operation—a non-negotiable redundancy for commercial clients like schools and surgeries.

By physically separating these water-side components into an indoor wall-hung box, the expensive outdoor unit is kept simpler and more compact. It also prevents the circulation pump and critical flow sensors from being exposed to freezing outdoor temperatures in the event of a total power failure.

SYSTEM INPUTS	SYSTEM OUTPUTS

Data Sources:
[LLM] Gemini 3.1 Pro — module decomposition

- Heated water/glycol from Outdoor Unit
  - 230-V AC Power

- Pressurized hot water flow to radiators / calorifiers
  - Flow-rate telemetry

Data Sources:

[LLM] Gemini 3.1 Pro — module decomposition

4.8.2 Specifications & Risk Profile

Expected Key Parts

- ErP-rated DC Circulation Pump
- Expansion Vessel (e.g., 12L)
- Inline Backup Immersion Heater
- Pressure Relief Valve (PRV)
- Vortex Flow Meter

Identified Failure Modes

- Circulation pump cavitation due to air trapped in the closed loop
- Expansion vessel membrane rupture causing massive pressure fluctuations
- Immersion heater scale build-up leading to element burnout

Open Questions

- Total hydraulic resistance of the legacy pipework in target retrofit buildings
- Vibration transmission of the pump into the internal building walls

Data Sources:
[LLM] Gemini 3.1 Pro — module decomposition

4.8.3 Cost Breakdown

No bill of materials rows allocated to this module yet.

Data Sources:
[LLM] Gemini 3.1 Pro — part generation
[Deterministic] Cost model calculation

4.9 Control Gateway & HMI

Lead Time: 14 weeks

Status: draft

Module Purpose

Provides user interaction, scheduling, and high-level BMS (Building Management System) integration.

Why it matters to the system

> Commercial buyers require BMS connectivity and easy diagnostics. Without professional-grade control integration, the product cannot compete in B2B HVAC tenders.

Detailed Technical Description

The Control Gateway & HMI is the singular touchpoint for facility managers and installers. Targeted squarely at the commercial sector, consumer-grade smart thermostats are insufficient. This interface provides granular visibility into flow temperatures, COP, compressor speeds, and fault codes. Its processing architecture analyzes indoor ambient metrics against the outdoor climatic conditions using advanced weather-compensation curves to demand specific water flow temperatures from the outdoor unit, maximizing ErP efficiency.

Crucially, this module serves as the cascade manager and BMS bridge. While the brief specifies a 30kW unit, commercial installations frequently require 60kW or 90kW setups, demanding two or three 30kW units acting in unison. The gateway possesses native master-slave logic, balancing runtime hours equally across cascaded outdoor units. It translates internal RS485 telemetry to standardized Modbus or BACnet, allowing integration into the building's existing central control networks.

To achieve the £4,500 system cost, developing a bespoke HMI from scratch is ruled out. The strategy utilizes a white-labeled, pre-certified commercial controller paired with a customized, brand-specific UI skin, vastly reducing software validation time and avoiding high NRE (Non-Recurring Engineering) expenses.

SYSTEM INPUTS	SYSTEM OUTPUTS

Data Sources:
[LLM] Gemini 3.1 Pro — module decomposition

- User setpoint inputs
- Room temperature sensor data
- BMS/Zone data via Modbus/BACnet

- Operating schedules and cascaded load demands
- System alerts and diagnostic codes

Data Sources:

[LLM] Gemini 3.1 Pro — module decomposition

4.9.2 Specifications & Risk Profile

Expected Key Parts

- Capacitive touch HMI LCD screen
- Communication gateway PCB (Modbus/BACnet/Wi-Fi)
- Wall-mounting bezel
- Local temperature/humidity sensor

Identified Failure Modes

- Corrupted firmware during over-the-air (OTA) updates bricking the interface
- RS-485 communication dropouts due to unshielded cables running near high voltage
- LCD backlight failure or touchscreen digitizer degradation over time

Open Questions

- Specific BMS protocols most desired by UK small commercial building owners
- Signal penetration efficacy for Wi-Fi/RF telemetry through reinforced concrete

Data Sources:

[LLM] Gemini 3.1 Pro — module decomposition

4.9.3 Cost Breakdown

No bill of materials rows allocated to this module yet.

Data Sources:
[LLM] Gemini 3.1 Pro — part generation
[Deterministic] Cost model calculation

5. Bill of Materials

No parts in bill of materials.

6. Cost Waterfall & Economics

UNIT COST	TARGET CEILING	HEADROOM
£0.00	**None**	**N/A**

Make vs Buy Split

PURCHASED (BUY) TOTAL	CUSTOM (MAKE) TOTAL	MAKE %
£0.00	**£0.00**	**0%**

Per-Module Roll-up

Per-module cost data not available.

Non-recurring engineering and overheads

Total non-recurring engineering (Tooling, Certs)	**£0.00**

Grand Total (Unit plus Non-recurring Engineering) £0.00

Data Sources:
[Deterministic] Domain overhead multiplier model

Cost Confidence Assessment

Costing methodology employs deterministic expansions over a historical parts library combined with spot-price indices for raw materials. Custom parts are calculated volumetrically and adjusted by domain-specific complexity multipliers to ensure realistic unit economics.

Make-versus-buy categorisation is automated based on process availability. Non-recurring engineering (NRE) totals reflect tooling, testing, and compliance costs amortised over the target batch size. Margins of error apply to COTS components subject to global supply chain volatility.

Data Sources:
[Deterministic] Domain overhead multiplier model

7.1 Compressor Assembly Risks

Risk 1: Refrigerant ignition at electrical terminals

Base Score: S:5 × L:1 = 5

CAUSE
Unknown

CONSEQUENCE
Unknown

EXISTING CONTROLS
None

PLANNED MITIGATION
Use only ATEX-certified, hermetically sealed compressor terminals.

Owner: Mechanical Design Lead

Risk 2: Excessive acoustic noise above MCS limits

Base Score: S:4 × L:3 = 12

CAUSE
Unknown

CONSEQUENCE
Unknown

EXISTING CONTROLS
None

PLANNED MITIGATION
Incorporate double-layered acoustic blanketing and custom anti-vibration rubber mounts.

Owner: Compliance Engineer

Data Sources:
[LLM] Gemini 3.1 Pro — FMEA analysis

7.1 Compressor Assembly Risks

Risk 3: Scroll fracture due to liquid slugging

Base Score: S:4 × L:2 = 8

CAUSE

Unknown

CONSEQUENCE

Unknown

EXISTING CONTROLS

None

PLANNED MITIGATION

Incorporate a high-volume suction accumulator in the refrigerant circuit.

Owner: Thermodynamics Engineer

Data Sources:
[LLM] Gemini 3.1 Pro — FMEA analysis

7.2 Evaporator Coil Module Risks

Risk 4: Frost-induced tube rupture

Base Score: S:4 × L:2 = 8

CAUSE
Unknown

CONSEQUENCE
Unknown

EXISTING CONTROLS
None

PLANNED MITIGATION
Optimize hot gas bypass routing to prioritize lower headers during defrost.

Owner: Thermodynamics Engineer

Risk 5: Galvanic corrosion between Cu and Al

Base Score: S:3 × L:3 = 9

CAUSE
Unknown

CONSEQUENCE
Unknown

EXISTING CONTROLS
None

PLANNED MITIGATION
Ensure robust epoxy coating on end plates and use high-grade hydrophilic barrier.

Owner: Materials Engineer

Data Sources:
[LLM] Gemini 3.1 Pro — FMEA analysis

7.2 Evaporator Coil Module Risks

Risk 6: Insufficient heat transfer area

Base Score: S:3 × L:2 = 6

CAUSE

Unknown

CONSEQUENCE

Unknown

EXISTING CONTROLS

None

PLANNED MITIGATION

Conduct CFD simulation and wind tunnel tests prior to final tooling lock.

Owner: Thermodynamics Engineer

Data Sources:
[LLM] Gemini 3.1 Pro — FMEA analysis

7.3 Air Movement Module Risks

Risk 7: Failing MCS noise test limits

Base Score: S:5 × L:3 = 15

CAUSE
Unknown

CONSEQUENCE
Unknown

EXISTING CONTROLS
None

PLANNED MITIGATION
Source premium COTS fans (e.g., ebm-papst or Ziehl-Abegg) with pre-certified acoustic data.

Owner: Compliance Engineer

Risk 8: Motor overheating at low speeds

Base Score: S:3 × L:2 = 6

CAUSE
Unknown

CONSEQUENCE
Unknown

EXISTING CONTROLS
None

PLANNED MITIGATION
Specify motors with built-in thermal protection embedded in stator windings.

Owner: Electrical Engineer

Data Sources:
[LLM] Gemini 3.1 Pro — FMEA analysis

7.3 Air Movement Module Risks

Risk 9: Ice build-up on fan blades causing imbalance

Base Score: S:4 × L:2 = 8

CAUSE

Unknown

CONSEQUENCE

Unknown

EXISTING CONTROLS

None

PLANNED MITIGATION

Implement fan-burst logic in software to clear moisture pre-freeze.

Owner: Control Systems Engineer

Data Sources:

[LLM] Gemini 3.1 Pro — FMEA analysis

7.4 Condenser BPHE Module Risks

Risk 10: Cross-contamination due to internal rupture

Base Score: S:5 × L:1 = 5

CAUSE
Unknown

CONSEQUENCE
Unknown

EXISTING CONTROLS
None

PLANNED MITIGATION
Ensure supplier provides UKCA/PESR burst test validation above 45 bar.

Owner: Manufacturing Engineer

Risk 11: Fouling drastically reducing COP

Base Score: S:3 × L:4 = 12

CAUSE
Unknown

CONSEQUENCE
Unknown

EXISTING CONTROLS
None

PLANNED MITIGATION
Mandate inline magnetic dirt separators and chemical dosing on the installer checklist.

Owner: Product Manager

Data Sources:
[LLM] Gemini 3.1 Pro — FMEA analysis

7.4 Condenser BPHE Module Risks

Risk 12: Energy loss to ambient environment

Base Score: S:2 × L:3 = 6

CAUSE

Unknown

CONSEQUENCE

Unknown

EXISTING CONTROLS

None

PLANNED MITIGATION

Design a tightly fitted custom closed-cell elastomeric insulation jacket.

Owner: Mechanical Design Lead

Data Sources:
[LLM] Gemini 3.1 Pro — FMEA analysis

7.5 Refrigerant Management Skid Risks

Risk 13: R290 slow leak via braze defects

Base Score: S:4 × L:3 = 12

CAUSE
Unknown

CONSEQUENCE
Unknown

EXISTING CONTROLS
None

PLANNED MITIGATION
Implement automated helium sniffer leak testing on 100% of production units.

Owner: Manufacturing Engineer

Risk 14: Liquid slugging damaging compressor

Base Score: S:4 × L:2 = 8

CAUSE
Unknown

CONSEQUENCE
Unknown

EXISTING CONTROLS
None

PLANNED MITIGATION
Oversize suction accumulator slightly and code aggressive superheat protection logic.

Owner: Control Systems Engineer

Data Sources:
[LLM] Gemini 3.1 Pro — FMEA analysis

7.6 Outdoor Electrical Enclosure Risks

Risk 15: Ignition of leaked R290 gas

Base Score: S:5 × L:1 = 5

CAUSE
Unknown

CONSEQUENCE
Unknown

EXISTING CONTROLS
None

PLANNED MITIGATION
Design enclosure for continuous positive pressure or complete ATEX intrinsic sealing.

Owner: Electrical Engineer

Risk 16: Inverter thermal throttling

Base Score: S:3 × L:3 = 9

CAUSE
Unknown

CONSEQUENCE
Unknown

EXISTING CONTROLS
None

PLANNED MITIGATION
Mount inverter heat sink directly in the main condenser fan airflow stream.

Owner: Mechanical Design Lead

Data Sources:
[LLM] Gemini 3.1 Pro — FMEA analysis

7.6 Outdoor Electrical Enclosure Risks

Risk 17: Grid compliance failure (G99)

Base Score: S:4 × L:2 = 8

CAUSE

Unknown

CONSEQUENCE

Unknown

EXISTING CONTROLS

None

PLANNED MITIGATION

Conduct early pre-compliance harmonics and flicker testing on Alpha prototypes.

Owner: Compliance Engineer

Data Sources:
[LLM] Gemini 3.1 Pro — FMEA analysis

7.7 Structural Chassis & Casing Risks

Risk 18: Panel rattle/resonance exceeding MCS noise limits

Base Score: S:4 × L:3 = 12

CAUSE
Unknown

CONSEQUENCE
Unknown

EXISTING CONTROLS
None

PLANNED MITIGATION
Use FEM analysis to optimize fold stiffness and add butyl rubber damping pads.

Owner: Mechanical Design Lead

Risk 19: Corrosion in coastal environments

Base Score: S:3 × L:2 = 6

CAUSE
Unknown

CONSEQUENCE
Unknown

EXISTING CONTROLS
None

PLANNED MITIGATION
Apply C4 grade marine powder coating as standard or as a premium upgrade.

Owner: Materials Engineer

Data Sources:
[LLM] Gemini 3.1 Pro — FMEA analysis

7.7 Structural Chassis & Casing Risks

Risk 20: Water pooling in base pan

Base Score: S:4 × L:3 = 12

CAUSE

Unknown

CONSEQUENCE

Unknown

EXISTING CONTROLS

None

PLANNED MITIGATION

Design aggressive pan sloping and integrate a low-wattage base pan heater trace.

Owner: Thermodynamics Engineer

Data Sources:
[LLM] Gemini 3.1 Pro — FMEA analysis

7.8 Indoor Hydrobox Module Risks

Risk 21: Inadequate flow rate triggering high-pressure trips

Base Score: S:4 × L:3 = 12

CAUSE
Unknown

CONSEQUENCE
Unknown

EXISTING CONTROLS
None

PLANNED MITIGATION
Select a high-head oversized DC pump capable of overcoming old, restrictive pipework.

Owner: Thermodynamics Engineer

Risk 22: Water leakage damaging indoor property

Base Score: S:3 × L:2 = 6

CAUSE
Unknown

CONSEQUENCE
Unknown

EXISTING CONTROLS
None

PLANNED MITIGATION
Integrate a water leak detection tray and automatic isolation valves inside the hydrobox.

Owner: Mechanical Design Lead

Data Sources:
[LLM] Gemini 3.1 Pro — FMEA analysis

7.9 Control Gateway & HMI Risks

Risk 23: Cybersecurity vulnerability via Wi-Fi module

Base Score: S:4 × L:2 = 8

CAUSE

Unknown

CONSEQUENCE

Unknown

EXISTING CONTROLS

None

PLANNED MITIGATION

Utilize end-to-end encryption and restrict remote control to secure VPN tunnels.

Owner: Control Systems Engineer

Risk 24: Cascade logic software bugs

Base Score: S:4 × L:3 = 12

CAUSE

Unknown

CONSEQUENCE

Unknown

EXISTING CONTROLS

None

PLANNED MITIGATION

License established cascade algorithms from a tier-1 controls vendor rather than building in-house.

Owner: Product Manager

Data Sources:

[LLM] Gemini 3.1 Pro — FMEA analysis

7.9 Control Gateway & HMI Risks

Risk 25: Installer wiring error destroying comms port

Base Score: S:3 × L:4 = 12

CAUSE
Unknown

CONSEQUENCE
Unknown

EXISTING CONTROLS
None

PLANNED MITIGATION
Implement opto-isolators and ruggedized over-voltage protection on all low-voltage terminals.

Owner: Electrical Engineer

Data Sources:
[LLM] Gemini 3.1 Pro — FMEA analysis

8. Supplier Shortlist

No supplier matches found during the search phase.

9. Audit Log

Pipeline execution log — generated by PDF Engine v2.1.

This document was generated automatically. Contents represent the synthesised output of the specialist pipeline.

Pipeline Execution Summary

Project Identifier	**a_compact_commercial_airsource_heat_pump**
Research Result	**Generated**
System Modules	**9 Modules**
Bill of Materials Lines	**0 Parts**
Cost Analysis	**Pending**
Specialist Reviews	**0 Completed**
Supplier Shortlists	**0 Matched**

10. Source Attribution

This table documents the origin of data for each major section of the report, ensuring traceability of LLM outputs versus deterministic or user-provided data.

SECTION	SOURCE	PROVIDER / SYSTEM	DETAIL
Research	LLM	OpenRouter	Gemini 3.1 Pro via OpenRouter
Research	USER	user	Founder brief text
Regulatory	LLM	Unknown LLM	Gemini 3.1 Pro — standards extraction
Modules	LLM	Unknown LLM	Gemini 3.1 Pro — module decomposition